

PPE Detection Model

Dataset Analysis and Equipment Identification Report

1. Introduction

This report presents an analysis of the personal protective equipment (PPE) found in the dataset used to train a YOLOv8-based detection model, examined within the framework of industrial standards, along with a class-level evaluation of model performance.

The dataset consists primarily of construction site imagery and contains 12 distinct classes. The model was trained using the YOLOv8n architecture over 50 epochs.

2. PPE Equipment in Dataset and Industrial Standards

The table below summarizes the technical industry names, international standard codes, and application areas of the PPE equipment identified in the dataset.

Equipment	Standard	Industry Name / Type	Color	Application Area
Helmet	EN 397	Industrial Hard Hat	White, Orange, Yellow	Construction site
Vest	EN ISO 20471:2013	Hi-Vis Warning Vest – Class 2	Fluorescent Red, Yellow	General construction, road
Gloves	EN 388	Cotton Knit Work Glove	White	Assembly, general work
Boots	EN ISO 20345 / 20347	Safety Boot / Rubber Wellington Boot	Black, Red	Construction, wet surfaces
Goggles	EN 166	Wrap-around Protective Eyewear	Clear / Tinted lens	General protection

3. Equipment Detail Descriptions

3.1 Helmet — EN 397 Industrial Hard Hat

The dataset predominantly features EN 397-compliant industrial hard hats with rigid plastic (ABS/HDPE) shells in white, orange, and yellow color variants. White helmets represent engineer/supervisor profiles, while orange and yellow helmets represent general construction workers.

3.2 Vest — EN ISO 20471 Class 2 High-Visibility Warning Vest

Class 2 high-visibility warning vests compliant with EN ISO 20471:2013 are predominantly observed in the dataset. Fluorescent red/pink and yellow-green color variants are present. They feature horizontal and diagonal retroreflective strips (yellow-green) in a sleeveless vest form.

3.3 Gloves — EN 388 Cotton Knit Work Glove

White cotton or cotton/polyester blend knit work gloves stand out as the dominant glove type in the dataset. These gloves, providing mid-level mechanical protection under EN 388, are widely used in general construction and assembly work.

3.4 Boots — EN ISO 20345 / EN ISO 20347

Two different boot types are observed in the dataset: black steel/composite toe safety boots (S1/S3) compliant with EN ISO 20345 form the dominant group, while red PVC rubber wellington boots under EN ISO 20347 are also present to a lesser extent.

3.5 Goggles — EN 166 Protective Eyewear

Wrap-around clear protective eyewear compliant with EN 166 is predominantly present in the dataset. Tinted filter (EN 172) sun-protective safety glasses are also found to a lesser degree. Clear lenses (filter code: 1) indicate indoor/sheltered construction site usage.

4. Model 1 — YOLOv8n Performance (12-Class, AP50 Scores)

The table below presents the AP50 scores achieved per class after 50 epochs of training with the YOLOv8n model on the original 12-class dataset.

Class	AP50 Score	Performance	Assessment
helmet	0.945	Very High	Sufficient and diverse samples in dataset
vest	0.963	Very High	Sufficient and diverse samples in dataset
boots	0.935	Very High	Sufficient and diverse samples in dataset
no_vest	0.735	Good	Model performs well
Person	0.710	Good	Model performs well
none	0.520	Moderate	Can be improved with additional data
gloves	0.499	Moderate	Can be improved with additional data
goggles	0.478	Moderate	Can be improved with additional data
no_helmet	0.304	Low	Insufficient samples, recommended to remove from model
no_goggle	0.400	Low	Can be improved with additional data
no_boots	0.090	Very Low	Insufficient samples, recommended to remove from model

Class	AP50 Score	Performance	Assessment
no_gloves	0.043	Very Low	Insufficient samples, recommended to remove from model

5. Model 2 — YOLO26n (8-Class, Null Samples)

Following the removal of the four low-performing 'no_' classes (no_helmet, no_goggle, no_gloves, no_boots), a second model was trained using the YOLO26n architecture — the latest member of the Ultralytics YOLO family released in January 2026. YOLO26n introduces NMS-free end-to-end inference, ProgLoss, STAL label assignment, and the MuSGD optimizer, making it particularly suited for edge deployment.

In this version, images that previously contained only 'none' class annotations were converted to unannotated negative samples, allowing the model to learn background suppression without a placeholder bounding box class.

Class	AP50 Score	Performance	Assessment
all (mAP50)	0.740	Good	Model performs well
Person	0.660	Good	Can be improved with additional data
boots	0.890	Very High	Model performs well
gloves	0.480	Moderate	Can be improved with additional data
goggles	0.510	Moderate	Can be improved with additional data
helmet	0.930	Very High	Sufficient and diverse samples in dataset
vest	0.960	Very High	Sufficient and diverse samples in dataset